

Verified Re-derivation of Pope's Two-Dimensional Effective-Viscosity Hypothesis

1 Objective and notation

This document independently reduces equation (4.2) of Pope (1975) to the explicit relations (4.3)–(4.4) and records the symbolic and numerical checks used in the reproduction. Braces in the paper denote a trace; here they are written $\text{tr}(\cdot)$. The nondimensional strain $\mathbf{s}$ is symmetric and traceless, whereas the nondimensional rotation $\boldsymbol{\omega}$ is antisymmetric. Both are planar but are embedded in three dimensions. Pope's implicit relation is

$$\mathbf{a} = -g \left[b_1 \mathbf{s} + b_2 \left(\mathbf{a}\mathbf{s} + \mathbf{s}\mathbf{a} - \frac{2}{3} \mathbf{I}_3 \text{tr}(\mathbf{a}\mathbf{s}) \right) - b_3 (\mathbf{a}\boldsymbol{\omega} - \boldsymbol{\omega}\mathbf{a}) \right], \quad (1)$$

where

$$b_1 = \frac{8}{15}, \quad b_2 = \frac{5 - 9C_2}{11}, \quad b_3 = \frac{7C_2 + 1}{11}, \quad g = \left(C_1 + \frac{P}{\epsilon} - 1 \right)^{-1}. \quad (2)$$

The numerical values used for Figure 1 are $C_1 = 1.5$ and $C_2 = 0.4$.

2 Planar tensor representation

Every planar symmetric traceless tensor and planar antisymmetric tensor can be written, in a coordinate system whose third direction is homogeneous, as

$$\mathbf{s} = \begin{pmatrix} p & q & 0 \\ q & -p & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \boldsymbol{\omega} = \begin{pmatrix} 0 & r & 0 \\ -r & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (3)$$

Consequently,

$$S_2 \equiv \text{tr}(\mathbf{s}^2) = 2(p^2 + q^2), \quad W_2 \equiv \text{tr}(\boldsymbol{\omega}^2) = -2r^2. \quad (4)$$

The three independent symmetric, traceless basis tensors are

$$\mathbf{T}^0 = \frac{1}{3} \mathbf{I}_3 - \frac{1}{2} \mathbf{I}_2, \quad \mathbf{T}^1 = \mathbf{s}, \quad \mathbf{T}^2 = \mathbf{s}\boldsymbol{\omega} - \boldsymbol{\omega}\mathbf{s}. \quad (5)$$

They are mutually orthogonal under the matrix inner product $\mathbf{X} : \mathbf{Y} = \text{tr}(\mathbf{X}\mathbf{Y})$. Thus write

$$\mathbf{a} = G^0 \mathbf{T}^0 + G^1 \mathbf{T}^1 + G^2 \mathbf{T}^2. \quad (6)$$

3 Reduction to three scalar equations

Substitution of (6) into (1), followed by projection of the residual onto each tensor in (5), gives

$$G^0 = 2b_2gS_2G^1, \quad (7)$$

$$G^1 = -g \left(b_1 - \frac{1}{3}b_2G^0 - 2b_3W_2G^2 \right), \quad (8)$$

$$G^2 = b_3gG^1. \quad (9)$$

For example, the direct matrix projection first yields $G^0 - 4b_2g(p^2 + q^2)G^1 = 0$; equation (7) follows from (4). This explicit parameterization avoids requiring a computer algebra system to infer tensor symmetries.

Substituting (7) and (9) into (8) and collecting G^1 gives

$$G^1 = -\frac{b_1g}{D}, \quad D = 1 - 2b_3^2g^2W_2 - \frac{2}{3}b_2^2g^2S_2. \quad (10)$$

Defining

$$C_\mu \equiv \frac{b_1g}{2D} = \frac{\frac{1}{2}b_1g}{1 - 2b_3^2g^2\text{tr}(\boldsymbol{\omega}^2) - \frac{2}{3}b_2^2g^2\text{tr}(\boldsymbol{s}^2)} \quad (11)$$

therefore gives

$$G^1 = -2C_\mu, \quad G^2 = -2C_\mu gb_3, \quad G^0 = -4C_\mu gb_2 \text{tr}(\boldsymbol{s}^2). \quad (12)$$

Finally, inserting (12) into (6) yields

$$\boxed{\boldsymbol{a} = -2C_\mu \left[\boldsymbol{s} + gb_3(\boldsymbol{s}\boldsymbol{\omega} - \boldsymbol{\omega}\boldsymbol{s}) + gb_2 \text{tr}(\boldsymbol{s}^2) \left(\frac{2}{3}\boldsymbol{I}_3 - \boldsymbol{I}_2 \right) \right]}. \quad (13)$$

Equations (13) and (11) reproduce Pope's equations (4.3) and (4.4), respectively.

4 Symbolic verification with SymPy

The script `symbolic_derivation.py` defines the general matrices in (3), constructs the basis and the residual of (1), and performs the three projections exactly. All rational numbers, especially $1/3$ and $2/3$, are represented with `sympy.Rational`, so no floating-point approximation enters the algebra. SymPy's `solve` function solves equations (7)–(9). The resulting expressions are substituted into the original 3×3 residual, after which `factor` and `cancel` reduce every matrix entry exactly to zero.

The symbolic calculation is a check rather than a hidden source of the result: the matrix parameterization, projected equations, scalar solution, and final substitution are all exposed by the script. The projection divides by basis norms for a generic nonzero strain and rotation; because the resulting identities are polynomial identities, the zero-strain and zero-rotation limits follow by continuity and are tested separately.

5 Reduction used for Figure 1

Pope defines

$$\sigma = \left(\frac{1}{2} \text{tr}(\boldsymbol{s}^2) \right)^{1/2}, \quad \Omega = \left(-\frac{1}{2} \text{tr}(\boldsymbol{\omega}^2) \right)^{1/2}. \quad (14)$$

Thus $S_2 = 2\sigma^2$ and $W_2 = -2\Omega^2$, and equation (11) becomes

$$C_\mu = \frac{\frac{1}{2}b_1g}{1 + 4b_3^2g^2\Omega^2 - \frac{4}{3}b_2^2g^2\sigma^2}. \quad (15)$$

For incompressible flow, the isotropic stress and rotation make no contribution to production, so

$$\frac{P}{\epsilon} = -\text{tr}(\mathbf{a}\mathbf{s}). \quad (16)$$

Contracting (13) with $\mathbf{s}$ shows that its second and third basis terms do not contribute, giving

$$\frac{P}{\epsilon} = 2C_\mu \text{tr}(\mathbf{s}^2) = 4C_\mu\sigma^2, \quad g^{-1} = C_1 - 1 + 4C_\mu\sigma^2. \quad (17)$$

The numerical solver evaluates equations (15) and (17) together. More specifically, with

$$A = C_1 - 1, \quad K = 4b_3^2\Omega^2 - \frac{4}{3}b_2^2\sigma^2, \quad (18)$$

it solves the single eliminated equation

$$(1 - Ag)(1 + Kg^2) - 2b_1\sigma^2g^2 = 0. \quad (19)$$

For $\sigma > 0$, the positive physical root lies in $0 < g < 1/A$ and is selected continuously from the zero-strain limit. At $\sigma = 0$, $g = 1/A$ is used directly. Equation (15) then supplies C_μ .

6 Verification strategy and result

Development followed short red–green–refactor cycles. The retained tests check: tensor symmetry, tracelessness, orthogonality, and analytical invariants; a direct linear solve of the implicit equation; rotational covariance; agreement of the explicit formula with that independent solve; exact symbolic substitution; the production identity; both coupled scalar equations; limiting behavior; and final PNG generation. Floating-point tolerances are at the scale expected for well-conditioned, small double-precision linear systems and for the stated root-solver tolerance.

These checks verify the algebra and its implementation. They do not validate the physical Reynolds-stress closure assumptions used to obtain (1).

Reproducibility

From this directory, run

```
python -m uv sync
python -m uv run pytest
python -m uv run python symbolic_derivation.py
python -m uv run python plot_figure1.py
latexmk -pdf -interaction=nonstopmode -halt-on-error derivation.tex
```

The Python environment is locked by `uv.lock`.

References

- [1] S. B. Pope, “A more general effective-viscosity hypothesis,” *Journal of Fluid Mechanics*, vol. 72, part 2, pp. 331–340, 1975.

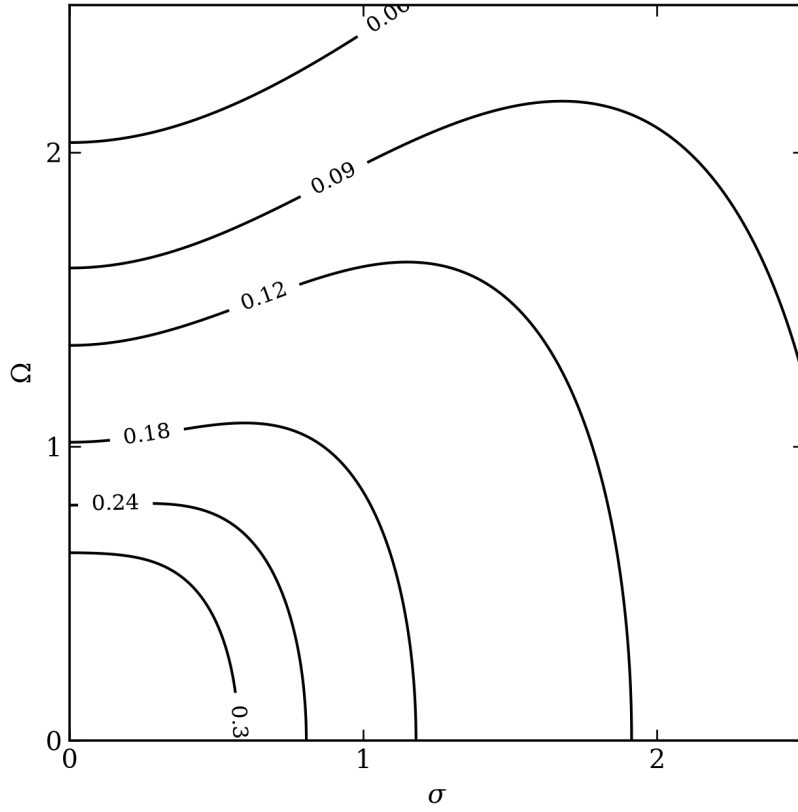


Figure 1: Reproduction of Pope (1975), Figure 1: contours of C_μ as a function of the strain invariant σ and rotation invariant Ω .